DESIGN OF AN ADAPTIVE CRUISE CONTROL SYSTEM USING PID CONTROL METHOD ON ELECTRIC VEHICLE PROTOTYPES

1st Joko Slamet Saputro
dept. Electrical Engineering
Universitas Sebelas Maret
Surakarta, Indonesia
jssaputro89@staff.uns.ac.id

2nd M. Shafiy Widi Kresno Wibisono
dept. Electrical Engineering
Universitas Sebelas Maret
Surakarta, Indonesia
wibishafy@student.uns.ac.id

3rd Joko Hariyono
dept. Electrical Engineering
Universitas Sebelas Maret
Surakarta, Indonesia
jokohariyono@staff.uns.ac.id

4th Miftahul Anwar
dept. Electrical Engineering
Universitas Sebelas Maret
Surakarta, Indonesia
miftahwar@staff.uns.ac.id

5th Warindi
dept. Electrical Engineering
Universitas Sebelas Maret
Surakarta, Indonesia
warindi@staff.uns.ac.id

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Abstract*—The use of Internal Combustion Engine (ICE) vehicles is still dominant in Indonesia due to easy access to petroleum-based fuels, but it has adverse effects on the environment. Vehicles are not only used for intra-city travel but also for long distances, increasing the risk of accidents as drivers may lose focus due to fatigue. To address this issue, environmentally friendly vehicles such as electric vehicles equipped with safety features to assist drivers are required. One such feature is Adaptive Cruise Control (ACC), which helps drivers reduce speed and stop the vehicle when approaching other vehicles to avoid collisions. This research focuses on implementing LiDAR sensors to detect the distance between vehicles in the Adaptive Cruise Control system using PID control method as an additional feature in an electric vehicle prototype. The ACC system uses an algorithm that automatically performs vehicle speed adjustment based on the distance detected by the sensors. The ACC system will work based on the given input speed and adjust the safe distance as well as the minimum distance from the given input speed. The TFMini-Plus LiDAR sensor is used, and the tests show that this sensor has a high accuracy of 98.43%, with an error value of 1.57%. The performance testing of the ACC system shows that it functions according to the specified algorithm. The system is also able to respond well to new objects, with an average rise time of less than 2 seconds, overshoot less than 3%, and settling time less than 12 seconds. However, there are several faults in the encoder's readings of speed changes, leading to suboptimal analysis of the system's response.

***Keywords*— Electric Vehicle, Adaptive Cruise Control, PID, Speed Adjustment, Safe Distance, System Response**

I. INTRODUCTION

Internal Combustion Engine (ICE) vehicles are a type of vehicle that is still often found and used by Indonesians today. This is because it is very easy to find filling stations to meet the needs of fuel oil. Badan Pusat Statistik noted that the number of motorized vehicles in Indonesia increases every year. In 2018, the number of motorized vehicles in Indonesia was 126,508,776. Then in 2019, the number increased to 133,617,012. In 2020, the number increased to 136,137,451 [1].

However, it should be noted that the use of Internal Combustion Engine (ICE) type vehicles has adverse impacts. Therefore, it is necessary to develop vehicles that are environmentally friendly and can support the needs of its users. Currently, a lot of research has been carried out related to the development of electric vehicles. In addition, there are already electric vehicles from various brands that are starting to be traded.

The use of electric vehicles is not only for inter-city, but also for long distances. This causes drivers to lose focus when experiencing fatigue while driving. Lack of focus from the driver will increase the risk of accidents to be high. Korps Lalu Lintas Kepolisian RI noted that in 2022, there were 197 traffic accidents that caused 304 victims. The victims consisted of 28 deaths, 22 serious injuries, and 254 minor injuries [2]. These traffic accidents are most commonly caused by drivers' lack of focus while driving their vehicles. Considering that in addition to environmentally friendly vehicles, people need a vehicle that has many features that can help drivers to reduce speed and stop the vehicle when approaching other vehicles to avoid collisions. The feature is Adaptive Cruise Control (ACC).

Cruise Control is a feature embedded in the vehicle that can automatically maintain the speed of the vehicle constantly. With this feature, pressing the gas pedal by the driver is no longer needed during driving when cruise control has been activated. The acceleration of the vehicle can be automatically regulated by cruise control when there is a change in the motion of the vehicle caused by external forces [3].

In this research, it is focused on applying the LiDAR sensor to detect the distance between vehicles in the adaptive cruise control system which will be used as an additional feature in the electric vehicle prototype.

II. TECHNICAL APPROACH

A. Adaptive Cruise Control

Currently, there are many features embedded in vehicles. The features are incorporated in the Advanced Driver Assistance System or ADAS. ADAS is an active assistance system in vehicles that can help drivers drive more safely and comfortably.

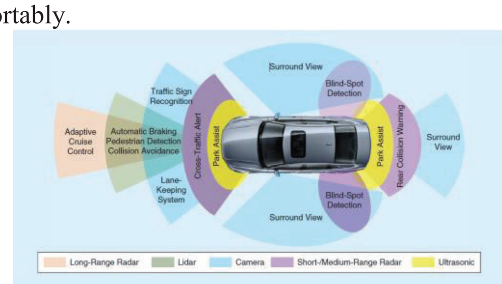

Fig. 1. Advanced Driver Assistance System

In the use of automated driving systems that use ADAS systems, there are several different levels. The levels start from level 0 to 5. The levels are set by SAE (Society of Automotive Engineers) in SAE J3016 [4].

B. Adaptive Cruise Control

Cruise Control is a technology embedded in automotive systems to keep the speed constant. This technology will adjust the acceleration and deceleration of the vehicle automatically when there is a change in motion due to external forces, such as friction [3].

ACC system allows the driver to maintain the desired driving speed if there is no vehicle in front of him and also maintain the desired distance to the vehicle in front. The ACC system can detect the distance to the vehicle in front of it using radar or LiDAR sensors. The distance information obtained by the sensor is used to generate commands to the throttle and brakes in order to adjust the speed and distance to the vehicle in front of it as predetermined [5].

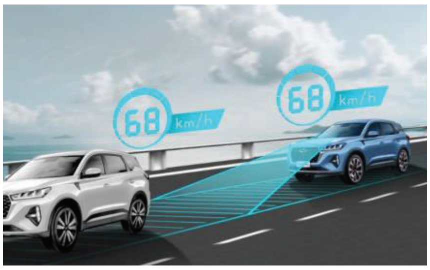

Fig. 2. Adaptive Cruise Control Illustration

III. Method

A. Algorithm

The algorithm specified in this adaptive cruise control system can automatically adjust the speed within a certain range. The driver will give input in the form of speed. Each speed value has its own safe distance and minimum distance. When the distance detected by the sensor is above the safe distance, the vehicle speed will be in accordance with the input given at the beginning. If the distance detected by the sensor is below the minimum distance, the speed will be 0. If the distance detected by the sensor is between the safe distance and the minimum distance, the system will adjust to reduce and increase the speed automatically. Automatic speed regulation is done using the following equation 2.1.

$$Speed = PID\ out \times \left(\frac{dist - min.dist}{safe.dist - min.dist}\right) \qquad (2.1)$$

B. PID Tuning Method

In this research, the control method used is PID control. This control method was chosen because it is the most commonly used method. To determine the most optimal PID parameters, a PID tuning method is required.

1) DC Motor Data Acquisition

DC motor data acquisition aims to determine the performance that can be generated by the motor used. The motor performance test data will be sent to the data acquisition software, PLX-DAQ, for further processing.

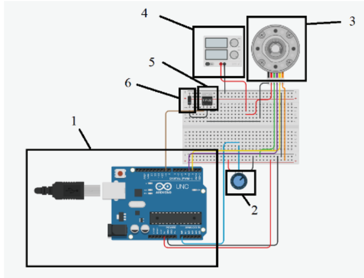

Fig. 3. DC Motor Data Acquisition Circuit

Figure 3 is a circuit used for motor data acquisition. Where the components used are Arduino Uno (1), Potentiometer (2), DC Motor Encoder (3), Power Supply (4), TIP 120 (5), and Diode (6).

2) Transfer Function

The method of determining the transfer function is done using the system identification toolbox in MATLAB. The dc motor data will be entered on the system identification. The data entered is input in the form of PWM and output in the form of RPM.

3) PID Tuning

PID control tuning is done using simulink which is a feature of MATLAB software and using the PID Tuner App on the PID Controller block. Figure shows the block diagram used to perform PID tuning.

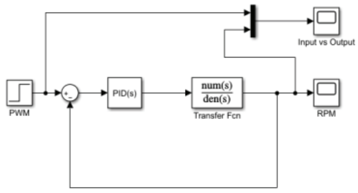

Fig. 4. PID Tuning Block Diagram

C. Component Assembly and Program Code Implementation

Figure 5 is the overall circuit of the adaptive cruise control system. The program code that has been made will then be implemented on the Raspberry Pi 4 Model B. Raspberry Pi will be connected to components such as the TFMini Plus sensor and BTS7960 motor driver. TFMini Plus is the main sensor of the system which is connected to Raspberry Pi using UART communication. The distance reading from the TFMini Plus sensor is input from the Raspberry Pi and will be used as a reference in the algorithm that has been made.

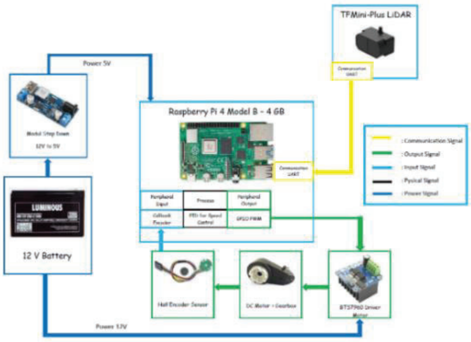

Fig. 5. Adaptive Cruise Control Circuit

BTS7960 is a driver that is connected to a DC motor and is used to control the motion of the DC motor. This motor driver gets its supply from the 12 V VRLA battery used in electric vehicles. The installed DC motor is connected to the Hall-Encoder Sensor to calculate the RPM of the DC motor which is used for the PID control method.

D. Encoder Calibration on DC Motor

Encoder sensor is a component that functions to measure the rotational speed produced by the motor (RPM) by counting the pulses generated. Encoder calibration aims to equalize the RPM reading value on the tachometer with the reading on the system.

Calculation of the RPM value performed by the encoder,

$$RPM = \frac{(Pulse\ Count \times 60\ detik)}{Pulse\ Per\ Revolution} \tag{2.2}$$

Calculation of RPM values on the motor shaft,

$$RPM\ Shaft = RPM \times Average\left(\frac{RPM\ Tachometer}{RPM\ Encoder}\right) \tag{2.3}$$

Calculation of Gear Ratio Value,

$$Gear\ Ratio = \frac{B}{A} \times \frac{D}{C} \times \frac{F}{E} \tag{2.4}$$

Calculation of RPM values on the wheels,

$$Wheel\ RPM = \frac{RPM\ Shaft}{Gear\ Ratio} \tag{2.5}$$

Calculation of real vehicle speed in km/hour,

$$Speed = \frac{(60 \times 2 \times \pi \times wheel\ rad \times wheel\ RPM)}{1000} \tag{2.6}$$

E. Speed Value Calibration Method

The speed value calibration method is a method to equalize the speed on vehicles in general and the speed on the vehicle used in the study. This method is done by measuring the maximum Shaft RPM value on the DC motor used using a tachometer. The maximum Shaft RPM value obtained is 19500 RPM. After that, the Wheel RPM calculation is carried out using equation 2.5 as follows.

$$Wheel\ RPM = \frac{RPM\ Shaft}{98{,}7}$$
$$Wheel\ RPM = 197{,}57$$

The Wheel RPM value is the calculation of RPM after passing through the gearbox and heading to the wheels on the electric vehicle prototype. The RPM value on the wheels obtained is 197.57 RPM. After getting the Wheel RPM value, the calculation can be converted to the speed value in km / hour using equation 2.6 as follows.

$$Speed = \frac{(60 \times 2 \times 3{,}14 \times 0{,}17 \times Wheel\ RPM)}{1000}$$
$$Kecepatan = 12.66\ \text{km/hour}$$

Calculation of the maximum speed value is done to determine the maximum speed of the electric vehicle prototype. Where after getting the maximum speed value, speed adjustments will be made to the vehicle used.

IV. Results and Discussion

A. Results

Components such as TFMini-Plus sensors, motor drivers, and encoders are connected to a double layer PCB that has been determined by the path to the pins on the Raspberry Pi 4. To connect from the double layer PCB that has been made to the Raspberry Pi 4, a pinout cable is used which connects all the pins on the Raspberry Pi to the PCB with just 1 cable.

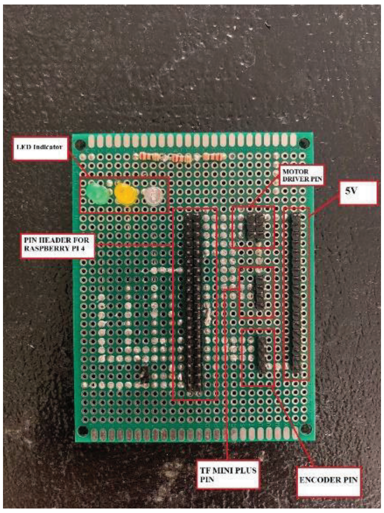

Fig. 6. Adaptive Cruise Control PCB

B. PID Design

1) Transfer Function

The transfer function is generated by giving 3 number of poles and 1 number of zeros (3rd order).

$$tf = \frac{88.93\,s + 204.1}{s^3 + 5.368\,s^2 + 9.318\,s + 2.262}$$

The transfer function obtained is 3rd order according to the provision of 3 poles and 1 zero. After obtaining the transfer function value, tuning will then be carried out to obtain PID parameters using simulink.

2) PID Tuning

The output graph obtained before using PID control is shown in Figure 7. It can be seen that before using PID control, the oscillations obtained are still numerous. Therefore, PID control is needed whose parameters can produce an optimal system response.

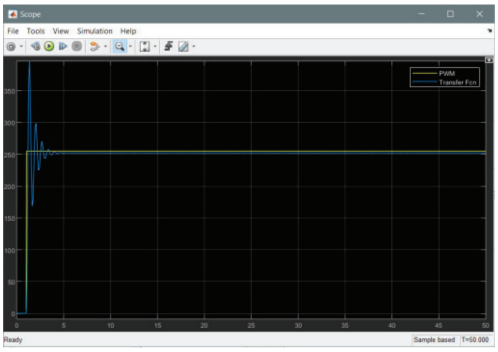

Fig. 7. System Response Without PID Control

PID parameter tuning is done using the PID Tuner App in the PID block in simulink. Tuning is done by adjusting the response time in the tuning tools in the PID

100	100 km/h	10 km/h	8 m	10 m

Tuner App then the author will manually change the PID parameters to get more optimal response parameters. The final PID parameters obtained after tuning and producing the most optimal response are as follows :

$$P = 0{,}047038$$
$$I = 0{,}021396$$
$$D = 0{,}015959$$

P = *Proportional*

I = *Integral*

D = *Derivative*

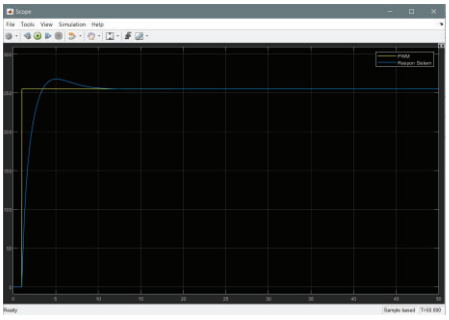

Fig. 8. System Response with PID Control

TABLE I. PID TUNING RESULTS

Rise Time	1,6 s
Settling Time	6,83 s
Overshoot	4,99%

Figure 8 is the result of the final tuning of the PID parameters. It can be seen that the oscillations that were previously present in the response have disappeared. There is still overshoot, but the value is small. Data on the final tuning results of PID parameters in simulink can be seen in table 4.4. The final results will then be applied to the adaptive cruise control system.

C. Performance Testing of Adaptive Cruise Control System

Table II shows the safe distance and minimum distance for each speed. The table has been modified from table provided by Wuling [6], because the maximum distance reading of the TFMini Plus sensor used only reaches 12 m. In addition, because the maximum speed of the car used in the study is 12 km/h, the maximum duty cycle sent is 100 which should produce 12 km/h will be adjusted to produce 120 km/h.

TABLE II. SPEED, SAFE DISTANCE, AND MINIMUM DISTANCE

Speed Input	Car Speed	Prototype Car Speed Adjustment	Minimum Distance	Safe Distance
30	30 km/h	3 km/h	1,5 m	3 m
40	40 km/h	4 km/h	2 m	4 m
50	50 km/h	5 km/h	2,5 m	5 m
60	60 km/h	6 km/h	4 m	6 m
70	70 km/h	7 km/h	5 m	7 m
80	80 km/h	8 km/h	6 m	8 m
90	90 km/h	9 km/h	7 m	9 m

D. Performance Testing with 100 km/h Input Speed

In this test, a speed input of 100 km/h was given. Where in this system it has been adjusted that the speed of 100 km/h at the input will produce 10 km/h on the car used in this study. With an input of 100 km/h, the vehicle has a safe distance of 10 meters and a minimum distance of 8 meters. In this test, it will be seen whether the vehicle can stop when the distance reading is 8 meters and reach the speed according to the input at a distance reading of 10 meters, and also will remain constant according to the input when the distance reading is above 10 meters according to the speed change algorithm that has been made.

1) Sensor Between Safe Distance and Minimum Distance

The object will be placed at points 800, 850, 900, 950, and 1000 cm. The speed value will be measured using 2 methods, namely by using an encoder that will appear on the system user interface and using direct measurement with a tachometer. The speed value displayed is in the form of RPM and speed in km/hour.

TABLE III. SPEED OUTPUT MEASURED BY ENCODER

Sensor Detection (cm)	Encoder	
	Speed (RPM)	Speed (KM/J)
1000	157,96	10,12
950	112,24	7,19
900	79,56	5,10
850	45,52	2,92
800	1,12	0,07

TABLE IV. SPEED OUTPUT MEASURED BY TACHOMETER

Sensor Detection (cm)	Tachometer (Real)	
	Speed (RPM)	Speed (KM/J)
1000	159,8	10,24
950	113	7,24
900	78,4	5,02
850	44,9	2,88
800	0	0

Tables III and IV present the results of the speed output reading by the encoder and measurement by the tachometer. When the sensor reading is at a distance of 800 cm, the resulting speed successfully reaches a value of ±0 km/hour. When the sensor reading is at a distance of 1000 cm, the resulting speed reaches ±10 km/hour. But in the results, there is still a difference or error resulting from the encoder and tachometer measurements.

TABLE V. SPEED ERROR

Sensor Detection (cm)	Error (%)	
	RPM	KM/J
1000	1,15	1,15

950	0,67	0,67
900	1,48	1,48
850	1,39	1,39
800	0	0
Average (%)	0,94	0,94

Table V presents the error value at each object placement point with a certain distance. The largest error value obtained is 1.15%. The average error value generated is 0.94%. From these results, it can be seen that the system has a high accuracy of ± 99%.

2) Sensor Detection Above Safe Distance

The object will be placed at points 1050, 1100, 1150, 1200, and 1250 cm. The speed value will also be measured using encoder and tachometer.

TABLE VI. SPEED OUTPUT MEASURED BY ENCODER

Sensor Detection (cm)	**Encoder**	
	Speed (RPM)	**Speed (KM/J)**
1050	158,77	10,17
1100	161,36	10,34
1150	160,59	10,29
1200	160,32	10,27
1250	161,71	10,36

TABLE VII. SPEED OUTPUT MEASURED BY TACHOMETER

Sensor Detection (cm)	**Tachometer (Real)**	
	Speed (RPM)	**Speed (KM/J)**
1050	159,2	10,20
1100	160,7	10,29
1150	160,6	10,29
1200	161,5	10,35
1250	161,3	10,33

Table VI and VII presents the results of the speed output reading by the encoder and the measurement by the tachometer. When the sensor reading is above a distance of 1000 cm, the resulting speed successfully reaches a value of ± 10 km/hour. However, in these results there is still a difference or error resulting from encoder and tachometer measurements.

TABLE VIII. SPEED ERROR

Sensor Detection (cm)	**Error (%)**	
	RPM	**KM/J**
1050	0,27	0,27
1100	0,41	0,41
1150	0,006	0,006
1200	0,73	0,73
1250	0,26	0,26
Average (%)	0,33	0,33

Table VIII presents the error value at each object placement point with a certain distance. The largest error value obtained is 0.73%. The average error value generated is 0.33%. From these results, it can be seen that the system has a high accuracy of ± 99%.

E. System Response When Detecting a New Object

1) Between Safe Distance and Minimum Distance

At this point, the system response is tested when detecting a new object between a safe distance and a minimum distance to determine the rise time, overshoot, and settling time values of the speed response of each detection change. Objects will be placed at points 800, 850, 900, 950, and 1000 cm. The object will be released alternately starting from point 800 to 1000 cm which causes the sensor to detect these distances continuously.

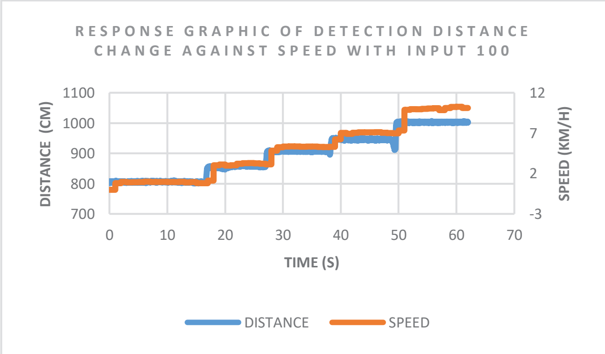

Fig. 9. System Response Graphic

TABLE IX. SYSTEM RESPONSE RESULTS

Distance (cm)	**Rise Time (s)**	**Overshoot (%)**	**Settling Time (s)**
850	1,33	2,38	9,33
900	1	1,45	10
950	1,9	1,52	10,8
1000	1,51	1,44	11,51
Average	1,44	1,70	10,41

The values of the response parameters are presented in table IX. From the test results that have been carried out, the average rise time value is 1.44 seconds. Meanwhile, the average overshoot value is 1.70%. The average settling time value obtained is 10.41 seconds.

2) Above Safe Distance

Objects will be placed at points 950, 1000, and 1050 cm. The object will be released alternately starting from point 950 to 1050 cm which causes the sensor to detect these distances continuously.

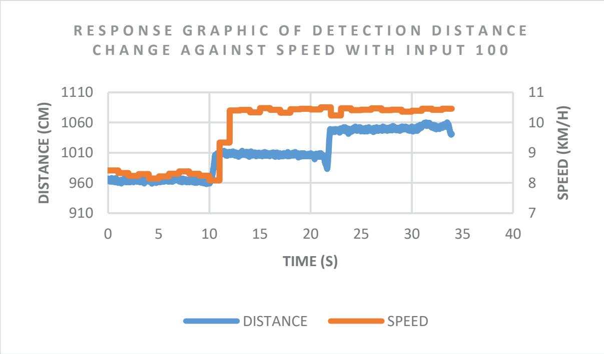

Fig. 10. System Response Graphic

TABLE X. SYSTEM RESPONSE RESULTS

Distance (cm)	Rise Time (s)	Overshoot / Fluctuation (%)	Settling Time (s)
1000	1,93	0,43	9,93
1050	-	0,45	11,25
Average	1,93	0,44	10,59

The values of the response parameters are presented in table X. However, the rise time value at some distance changes cannot be determined. That is because the distance is above the safe distance which causes the speed to have approximately the same value. From the test results that have been carried out, the average rise time value is 1.93 seconds. Meanwhile, the average overshoot/fluctuation value is 0.44%. The average settling time value obtained is 10.59 seconds.

F. *Testing Changes in Distance Over Speed*

In this test, the object will be moved slowly to see the speed change based on the object detection distance. Figure 11 is the result of the test that has been done. Testing is carried out for ± 30 seconds.

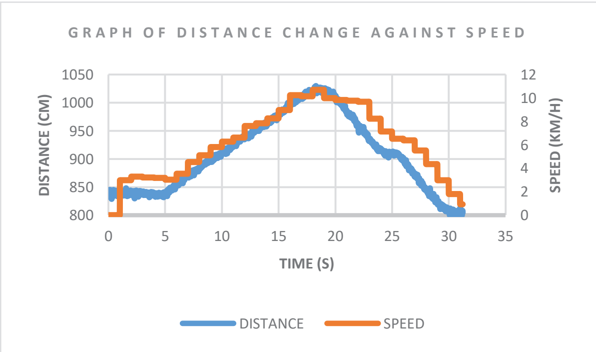

Fig. 11. Distance Change Against Speed Graphic

From the results it can be seen that changes in speed occur slowly based on changes in distance. However, due to the reading of the speed change value by the encoder which is not too detailed, causing there to be fractures in the resulting speed change reading.

V. CONCLUSION

The adaptive cruise control system was successfully designed and works according to the determined algorithm. The PID parameters obtained from the tuning results are P = 0.047038, I = 0.021396, and D = 0.015959. From the design of this PID control method, the results obtained rise time of 1.6 seconds, settling time of 6.83 seconds, and overshoot of 4.99%.

Performance testing is done with 2 methods, by testing each specific point and system response when detecting new objects. The test results at each point have proven that the system has run according to the predetermined algorithm. Where when the sensor detects a distance of more than a safe distance, the vehicle will continue to have a constant speed according to the input given. When the sensor detects the distance is at a minimum distance, the speed will be 0. In the test results of the system response when detecting a new object, the system gets an average rise time value of less than 2 seconds, an average overshoot of less than 3%, and an average settling time of less than 12 seconds. In testing a slow change in distance, it can be seen that changes in speed occur slowly based on changes in distance. The reading of the speed change value by the encoder which is not too detailed causes the analysis of the system response to be not optimal. It can be seen in the results that there are fractures in the resulting speed change readings.

VI. ACKNOWLEDGEMENT

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